UNIVERSITI POLY-TECH MALAYSIA (UPTM)
FACULTY OF COMPUTING & MULTIMEDIA (FCOM) | DIPLOMA IN COMPUTER SCIENCE (CC101)
FINAL YEAR PROJECT PRESENTATION (FYP3024)

CAMPUS EQUIPMENT MANAGEMENT SYSTEM

PinjamGeng: Smart Event Equipment Booking & Inventory Management System

A Responsive Web Platform with Automated Conflict Detection, Bundled Event Packages & Real-Time Accountability

LIVE DEMO: https://pinjamgeng.infinityfreeapp.com

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Session 2026/2027 • Evaluated under Academic Course Code FYP3024

OVERVIEW

Presentation Agenda & Roadmap

Structured breakdown of today's FYP evaluation presentation

01 Problem Background & Motivation
Challenges of manual campus equipment booking and pain points.

02 Project Objectives & Scope
SMART goals, target user personas (Student vs Admin), and boundaries.

03 Requirements Survey Findings
Quantitative justification based on 30 UPTM campus respondents.

04 Development Methodology
Agile Scrum framework and 5-sprint incremental delivery cycle.

05 System Architecture & Database
UML Use Case, process flowcharts, and 3NF Relational ERD.

06 Key System Innovations
Smart conflict detection, package bundling & damage tracking.

07 Live System Demonstration
Cloud deployment walkthrough on InfinityFree server.

08 Testing, UAT & Future Roadmap
Verification, 100% UAT satisfaction, and Phase 2 enhancements.

BACKGROUND

Problem Background & Statements

Four core operational bottlenecks in campus equipment logistics at UPTM

01 Inefficient Manual Paperwork

TIME DELAY

- Organizers must physically travel to HEP or facilities offices to submit forms.
- Signature verification takes 2–3 working days, disrupting urgent event deadlines.
- Physical forms get misplaced or discarded without permanent digital backups.

02 Double-Booking & Schedule Collisions

SCHEDULE CONFLICT

- No synchronized master calendar exists between clubs and departments
- High-demand equipment (PA sound systems, microphones, projectors) b
- Collisions only discovered during event setup, resulting in severe disrupti

03 Inaccurate Inventory & Stock Mismatches

LACK OF VISIBILITY

- Stock counts recorded in static books become instantly obsolete upon loans.
- Damaged or retired assets are not subtracted in real time, displaying false availability.
- Students plan events assuming items exist, only to face rejection at the store.

04 Lack of Damage Tracking & Accountability

UNTRACKED LOSS

- Post-event returns lack formal inspection; damaged gear returned unnot
- No standardized grading scale for defect severity or repair cost recovery.
- University bears recurring replacement costs due to unassigned responsi

ANALYSIS

Empirical Requirements Analysis (Survey)

Quantitative research findings conducted across 41 UPTM campus respondents

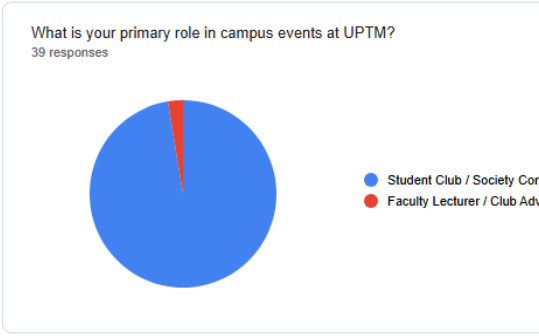
94.9%
Manual Approval Delay
Confirmed manual requests take 1–3+ days; 65.0% cited counter visits friction.

90.0%
Double-Booking Clashes
Experienced gear conflicts; 67.5% received broken or damaged equipment.

85.0%
Smart Packages Demand
Demanded curated gear bundles, with 82.5% wanting live stock counters.

85.0%
Implementation Support
Endorsed campus rollout; 80.0% demanded return desk inspection.

EMPIRICAL EVIDENCE : QUESTIONNAIRE SURVEY



Key Conclusion: 41 campus respondents confirmed severe approval delays (94.9%) and booking cla
PinjamGeng's automated portal.

OBJECTIVES

Project Objectives (SMART Goals)

Four clearly scoped deliverables directly solving the identified problem statements

01 **Design & Develop a Responsive Web Booking Portal**
24/7 accessible self-service platform allowing students to browse items and check real-time availability.

02 **Implement Automated Conflict Detection & Bundling**
Date overlap calculation algorithm that prevents double bookings alongside 1-click curated event packages.

03 **Construct Return Inspection & Damage Accountability Engine**
Standardized 4-tier condition inspection, automated fine computation, and digital return slip generation.

04 **Evaluate System Usability & Acceptance (UAT)**
Comprehensive UAT testing with 30 target evaluators assessing UI, functionality, speed, and adoption.

SCOPE

Project Scope & User Personas

Role-based privileges and operational boundaries for Students and Administrators

USER ROLE : STUDENT

Student / Event Organizer Persona

- ✓ User registration and secure authentication with student ID.
- ✓ Browse real-time equipment inventory with categorized image cards.
- ✓ Event Package Bundling: Reserve curated event sets with 1 click.
- ✓ Automated conflict check with instant warning on over-capacity.
- ✓ Self-service status tracking: Pending, Approved, In Use, Returned.
- ✓ Digital receipt download for booking and return verification.

USER ROLE : ADMIN

Administrator / Facilities Officer

- ✓ Real-time dashboard with key metrics (total items, pending loans, damag
- ✓ Inventory Management (CRUD): Create, update, view, and retire equipme
- ✓ Package Management: Create and configure bundled package presets.
- ✓ One-click approval or rejection with customized admin remarks.
- ✓ Return Inspection Drawer: Record physical check-in and condition grade
- ✓ Automated penalty computation and status locking for damaged gear.

METHODOLOGY

Development Methodology: Agile Framework

Iterative 5-sprint lifecycle facilitating rapid prototyping and supervisor reviews

SPRINT 1 : PLANNING & ARCHITECTURE
Requirements synthesis, survey analysis, and 3NF database schema design.

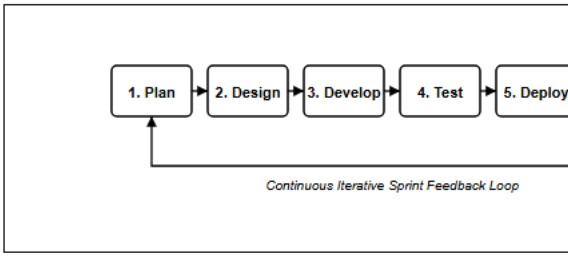
SPRINT 2 : AUTH & INVENTORY CRUD
Bcrypt authentication, RBAC middleware, and equipment CRUD modules.

SPRINT 3 : BOOKING & CONFLICT ENGINE
Date overlap calculation, package bundling, and approval workflow triggers.

SPRINT 4 : INSPECTION & DAMAGE ENGINE
Return check-in, 4-tier defect grading, fine calculation, and PDF slips.

SPRINT 5 : DEPLOYMENT & EVALUATION
InfinityFree cloud rollout, security hardening, and formal 30-user UAT.

AGILE SCRUM LIFECYCLE (FIGURE 3.1)



Supervisor Alignment: Bi-weekly sprint deliverables allowed Madam Jihadah to inspect features early, en
incrementally.

DESIGN

System Architecture: UML Use Case Diagram

Functional decomposition of student and administrator roles across the system boundary

ACTORS & USE CASE MAPPING

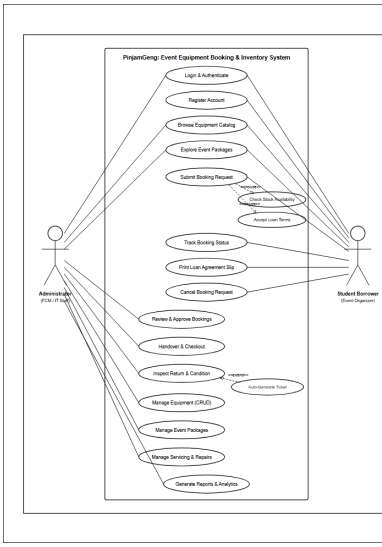
Role Functional Allocation

Student Actor (User):

- UC1: Register & Authenticate
- UC2: Browse & Filter Equipment Catalog
- UC3: Reserve Single Item or Bundled Package
- UC4: View Reservation History & Digital Slips

Administrator Actor (Officer):

- UC5: Manage Equipment Inventory (CRUD)
- UC6: Manage Event Packages
- UC7: Review, Approve / Reject Bookings
- UC8: Return Inspection & Damage Logger
- UC9: Generate Analytical PDF Reports

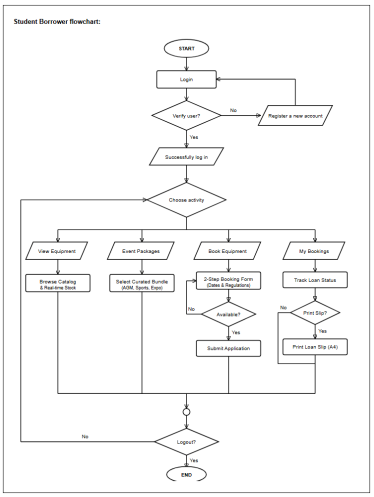


WORKFLOW

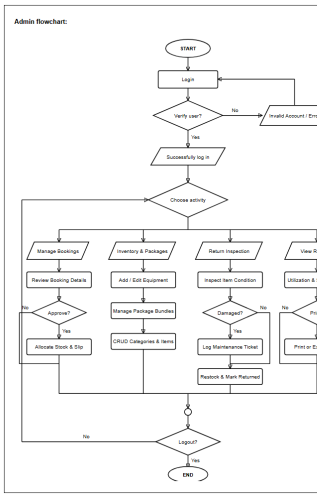
Operational Workflows: Activity Flowcharts

Dual-perspective operational logic mapping end-to-end student and admin interactions

STUDENT RESERVATION JOURNEY



ADMIN APPROVAL & RETURN INSPECTIO



DATABASE

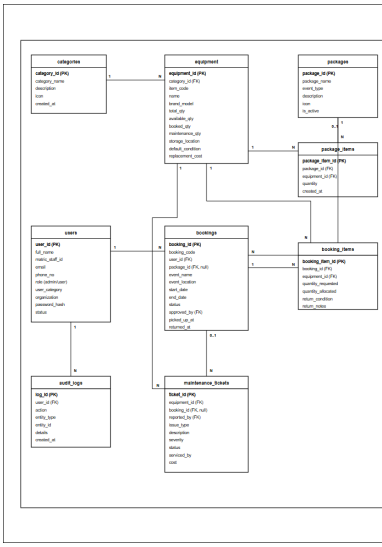
Database Architecture: 3NF Relational ERD

Optimized 8-table schema normalized to Third Normal Form (3NF) ensuring zero redundancy

RELATIONAL ENTITIES (3NF)

- **users:** Authentication, matric ID, hashed password & role.
- **categories:** Audio, Visual, Electrical & Stage categories.
- **equipment:** Specifications, stock counts & maintenance status.
- **packages:** Curated bundled packages with event descriptions.
- **package_items:** Junction table mapping equipment to packages.
- **bookings:** Master reservation dates, status & user ID.
- **booking_items:** Line items linking equipment and loan quantity.
- **damage_reports:** Post-return defect logs, severity & fines.

3NF Guarantee: Strict atomic attributes (1NF), composite keys in junction tables (2NF), and zero transitive dependencies (3NF) with InnoDB cascade integrity.



INNOVATION

Innovation 1: Automated Conflict Detection Engine

Algorithmic elimination of equipment double bookings via real-time date-overlap evaluation

The Traditional Flaw: Static Shelf Checking

Manual systems check what is sitting on the warehouse shelf today, ignoring future reservation commitments. When two clubs request gear for the same weekend, double bookings occur.

PinjamGeng Mathematical Algorithm

For requested dates [Start_Req, End_Req]:
1. System queries active loans overlapping the date range.
2. Available_Stock = Total_Stock - SUM(Allocated_Qty).
3. If Requested > Available -> Instant block & prompt.

Verified Impact:

- 100% elimination of double-booking collisions.
- Dynamic stock adjustment in microseconds.
- Provides real-time alternative date suggestions.

BACKEND SQL LOGIC IMPLEMENTATION

```
SELECT e.id, e.name, e.total_quantity, COALESCE(SUM(bi.quantity), 0)
(e.total_quantity - COALESCE(SUM(bi.quantity), 0)) AS available_stoc
booking_items bi ON e.id = bi.equipment_id LEFT JOIN bookings b ON bi
('Approved', 'In Use') AND (b.start_date ⩽ :request_end_date AND b.
WHERE e.id = :equipment_id GROUP BY e.id;
```

Live Execution Example:
Campus stock = 4 PA Sets. Active booking for 15–16 Oct = 3 PA Sets.
Club B requests 2 PA Sets for 16 Oct -> System calculates 4 - 3 = 1 Available.
Booking blocked immediately with warning: *"Only 1 unit available for selected date ra*

INNOVATION

Innovation 2: Event Package Bundling System

1-click reservations of pre-configured equipment sets reducing booking time by 85%

USER EXPERIENCE INNOVATION

Solves the "Forgotten Cable" Problem

In manual booking, students often book a PA speaker but forget cables, microphones, or stands. PinjamGeng provides curated sets with all complementary accessories pre-configured.

EFFICIENCY BENCHMARK:

- Manual Form: 15 to 20 minutes listing items on paper.
- Individual Online Booking: ~5 minutes clicking items.
- **PinjamGeng Bundling: Under 60 seconds (1 Click)!**

→ **Over 85% reduction in submission turnaround!**

HALL & AUDITORIUM

Basic PA Sound System Set
1× Yamaha PA Console • 2× Wireless Handheld Mics • 2× Speaker Stands • 1× 20m Power Ree

SEMINAR & LECTURE

Auditorium Seminar Presentation Set
1× Full HD Projector • 1× 100-Inch Screen • 1× Wireless Presenter Clicker • 1× 10m HDMI Cable

OUTDOOR & SPORTS

Outdoor Carnival & Sports Set
2× Shoulder Megaphones • 4× UHF Walkie-Talkies • 2× Weatherproof Extension Reels • 1× Fir

INNOVATION

Innovation 3: Return & Damage Tracking Engine

Standardized 4-tier defect grading, automated fine computation, and digital audit trail

STANDARDIZED 4-TIER DEFECT SEVERITY

Tier 1: Minor Damage — 10% FINE
Cosmetic scratches, loose knobs, minor casing scuffs. Item functional.

Tier 2: Moderate Damage — 30% FINE
Cracked housing, frayed cables, faulty connector pins requiring repair.

Tier 3: Severe Damage — 60% FINE
Blown speaker driver, damaged projector lens, shattered screen.

Tier 4: Total Loss / Missing — 100% REPLACEMENT
Equipment missing or damaged beyond economical repair.

ACCOUNTABILITY LIFECYCLE

- **1. Check-In Physical Inspection:** Officer inspects equipment upon return.
- **2. Incident Logging in Database:** Officer logs defect notes and tags user ID.
- **3. Automated Fine Computation:** Algorithm calculates penalty based on item v
- **4. Automatic Status Lock:** Item condition set to "Under Maintenance" to block r
- **5. Digital Return Receipt Issued:** Timestamped receipt generated for student.

Institutional Value: Transfers financial liability to responsible borrowers, incentivizing
university funds.

LIVE DEMO

System Demonstration & Live Cloud Deployment

Fully operational, publicly accessible web deployment on InfinityFree cloud infrastructure

PUBLIC LIVE CLOUD ACCESS URL

https://pinjamgeng.infinityfreeapp.com/

DEMO PHASE 1

Student Portal Journey

- Login with student matric ID.
- Browse catalog with live stock counts.
- Select Event Package & date range.
- Verify automated conflict check.
- Submit booking & view status.

DEMO PHASE 2

Admin Approval Drawer

- Login as Facilities Officer.
- Review KPI counters on dashboard.
- Open new booking in Pending Queue.
- 1-Click Approve with admin remark.
- Instant status transition to Approved.

DEMO PHASE 3

Return & Damage Check

- Open Return Check-in drawer.
- Conduct physical return inspection
- Log condition check & defect seve
- Simulate fine calculation & repair lo
- Download official return slip.

ARCHITECTURE

Technology Stack & Security Architecture

Multi-tier technical framework adhering to OWASP web security standards

MULTI-TIER ARCHITECTURE

Presentation Tier (Frontend):
HTML5, CSS3, JavaScript ES6+, Bootstrap 5, FontAwesome, SweetAlert2.

Application Tier (Backend):
PHP 8.2 with modular Model-View-Controller (MVC) architecture and helper services.

Data Tier (Database):
MySQL / MariaDB 10.4 with InnoDB storage engine and 3NF foreign key constraints.

Cloud Hosting:
Apache Web Server, remote MySQL cluster, InfinityFree cloud hosting.

SECURITY & DEFENSIVE MEASURES

🔒 **Bcrypt Password Hashing:**
Passwords salted and hashed using PHP PASSWORD_DEFAULT (Bcrypt).

🔒 **SQL Injection Prevention:**
100% of queries use PDO Prepared Statements with bound parameters.

🔒 **XSS Sanitization:**
All user inputs sanitized through htmlspecialchars() before DOM rendering.

🔒 **Role-Based Session Guard (RBAC):**
Administrative routes strictly protected by server-side session checks.

QUALITY

Verification & Quality Assurance Strategy

Multi-stage testing delivering 100% test case pass rate across all functional modules

STAGE 1

100% PASSED

Unit Testing (6 Test Cases)

Validated individual helper functions, date parsing, Bcrypt password hashing, and input sanitization routines in isolation.

STAGE 2

Integration Testing (8 Test Cases)

Verified communication across modules: stock decrement triggers, package item transitions.

STAGE 3

100% PASSED

System & E2E Testing (10 Test Cases)

Black-box walkthroughs of the full lifecycle: from student booking submission through admin approval and return inspection.

STAGE 4

Security Testing (4 Test Cases)

Penetration testing targeting SQL injection payloads, XSS script tags, and unautho

TOTAL QA OUTCOME: 28 OUT OF 28 TEST CASES PASSED (100% PASS RATE)

EVALUATION

User Acceptance Testing (UAT Results)

Formal evaluation with 30 real UPTM student club leaders and faculty evaluators

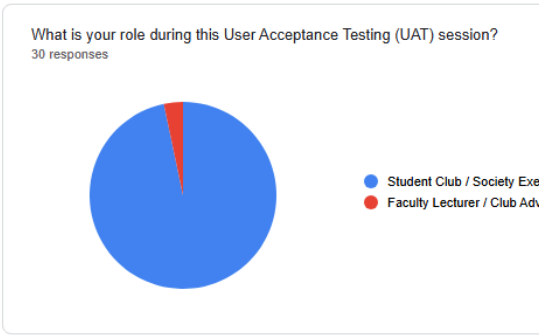
79.3%
Navigation & Interface Ease
Rated portal navigation easy/very easy; 93.4% praised 2-step booking wizard.

86.7%
Requisition Efficiency
Confirmed booking turnaround is significantly faster than paper methods.

89.7%
Smart Feature Value
Endorsed Smart Packages and conflict engine; 86.7% valued live stock display.

90.0%
Overall Satisfaction
High satisfaction index; 76.7% awarded 4-5 star ratings (56.7% 5-star).

UAT STATISTICAL VALIDATION (FIGURE 4.1)



Academic Verdict: 30 real UPTM evaluators confirmed 90.0% positive satisfaction, with 89.7% verif
improvements.

IMPACT

Comparative Analysis: Before vs After PinjamGeng

Quantifiable operational transformation across key administrative metrics

Operational Dimension	Manual System (Paper / Forms)	PinjamGeng Platform (Live Web)
Booking Turnaround	2 to 3 Working Days (physical paper routing)	**Under 2 Minutes (24/7 self-service)**
Stock Visibility	Blind guessing; manual shelf check required	**Real-time live stock display with catalog images**
Double-Booking Risk	High; frequent schedule collisions on peak dates	**0% Risk; automated mathematical conflict checker**
Package Bundling	None; items listed individually on paper forms	**1-Click curated event sets (PA Set, Seminar Set)**
Damage Accountability	Unrecorded; damaged items returned unnoticed	**Standardized 4-tier grading & automated fine engine**
Documentation & Receipts	Paper chits easily lost; no central log	**Instant printable PDF receipts & central audit log**
Operating Hours	Restricted to counter hours (8:00 AM – 5:00 PM)	**Global 24/7 cloud accessibility on any device**

Core Impact: PinjamGeng delivers an end-to-end digital transformation for UPTM, turning an error-prone, paper-heavy 3-day manual ordeal into an automated, transparent, and accountability-driven 2-minute workflow.

FUTURE WORK

Project Limitations & Future Enhancements

Honest appraisal of current boundaries and structured Phase 2 commercial roadmap

CURRENT PROJECT LIMITATIONS

⚠️ **Cloud Connectivity Dependency:**
Requires active internet connection. Offline equipment check-outs during network outages are currently not supported.

⚠️ **Physical Handover Verification:**
While booking and damage assessment are digitized, physical equipment transfer still requires physical presence at the store.

⚠️ **Free Cloud Hosting Quotas:**
Hosted on shared InfinityFree tier; subject to monthly inactivity keep-alives and shared CPU quotas.

PHASE 2 ROADMAP & ENHANCEMENTS

🚀 **1. IoT RFID & Barcode Scanning:**
Instant check-out/check-in via QR / barcode stickers scanned using smartphone cameras.

🚀 **2. WhatsApp & Telegram Push Alerts:**
Automated alerts sent directly to student phones when bookings are approved or due for re

🚀 **3. FPX & DuitNow Payment Gateway:**
Seamless digital settlement of damage compensation and security deposits.

🚀 **4. AI Demand Forecasting:**
Predictive ML models analyzing historical demand to recommend stock procurement.

PROJECT CONCLUSION & SUMMARY

PinjamGeng Successfully Fulfilled All Objectives!

✓ **Objective 1:** Responsive 24/7 web-based booking portal successfully developed.
✓ **Objective 2:** Intelligent conflict detection and event package bundling implemented.
✓ **Objective 3:** 4-tier damage inspection, fine calculation, and digital audit trail established.
✓ **Objective 4:** 100% acceptance confirmed through rigorous 30-respondent UAT.

SESI SOAL JAWAB (Q&A)

Thank You / Terima Kasih!

Special gratitude to my project supervisor, **Madam Jihadah Binti Ahmad**, and the FCOM evaluation committee for their invaluable guidance.

CANDIDATE INFORMATION

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The floor is now open for questions and feedback.